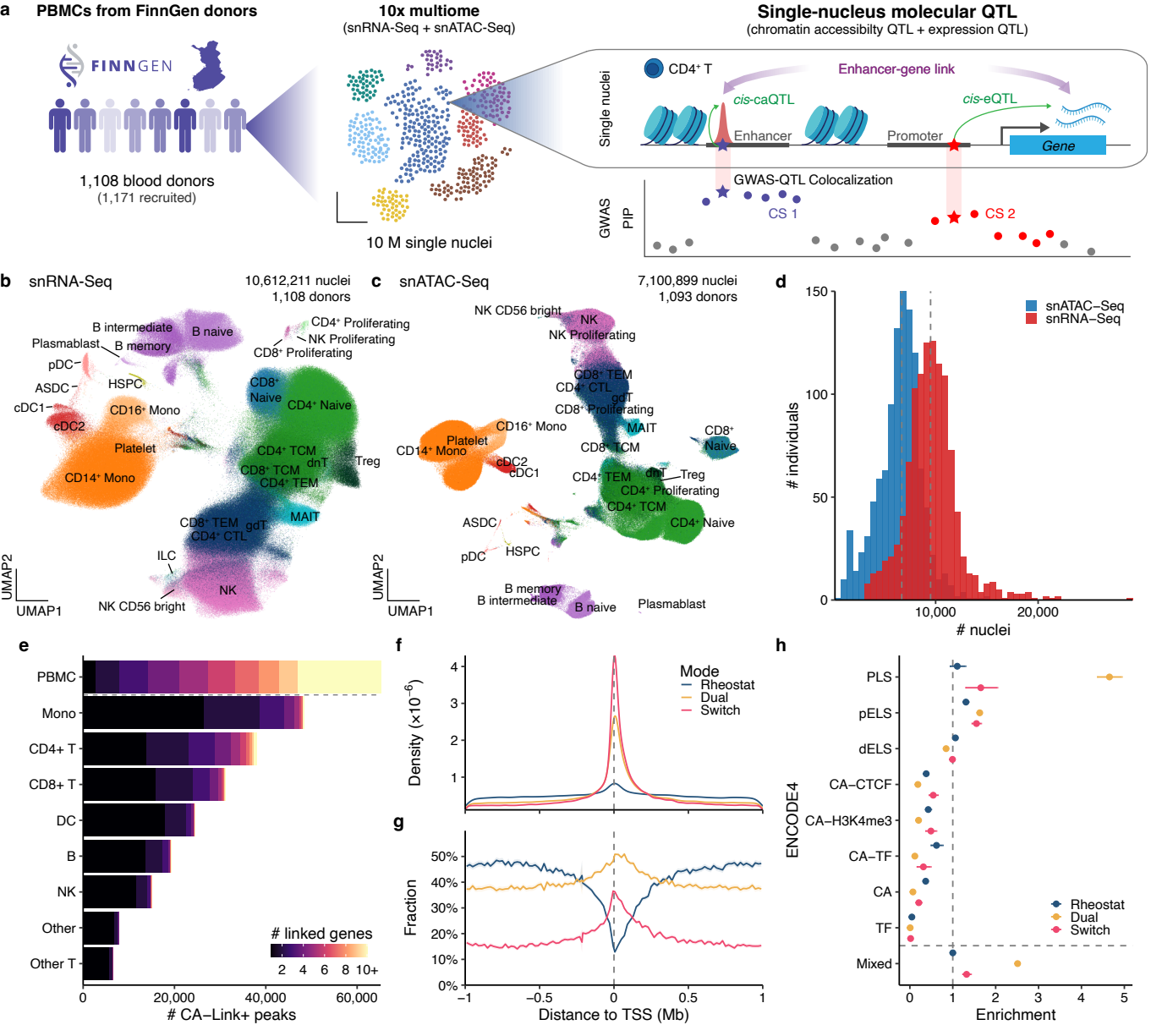
a
PBMCs from FinnGen donors
FINNGEN
1,108 blood donors
(1,171 recruited)
10x multiome
(snRNA-Seq + snATAC-Seq)
10 M single nuclei
Single-nucleus molecular QTL
(chromatin accessibilty QTL + expression QTL)
Single nuclei
CD4+ T
Enhancer-gene link
cis-caQTL
cis-eQTL
Enhancer
Promoter
Gene
GWAS-QTL Colocalization
GWAS
PIP
CS 1
CS 2
b
snRNA-Seq
10,612,211 nuclei
1,108 donors
B intermediate
B naive
Plasmablast
B memory
pDC
ASDC
HSPC
cDC1
cDC2
CD16+ Mono
Platelet
CD14+ Mono
CD4+ Proliferating
NK Proliferating
CD8+ Proliferating
CD8+ Naive
CD4+ Naive
CD4+ TCM
CD8+ TCM
dnT
Treg
CD4+ TEM
CD8+ TEM
gdT
CD4+ CTL
MAIT
ILC
NK
NK CD56 bright
UMAP2
UMAP1
c
snATAC-Seq
7,100,899 nuclei
1,093 donors
NK CD56 bright
NK
NK Proliferating
CD8+ TEM
CD4+ CTL
gdT
CD8+ Proliferating
CD16+ Mono
MAIT
CD8+ Naive
Platelet
CD14+ Mono
CD8+ TCM
cDC2
cDC1
CD4+ TEM
dnT
Treg
CD4+ Proliferating
CD4+ TCM
ASDC
CD4+ Naive
pDC
HSPC
B memory
B intermediate
B naive
Plasmablast
UMAP2
UMAP1
d
snATAC−Seq
snRNA−Seq
individuals
nuclei
150
100
50
0
10,000
20,000
e
PBMC
Mono
CD4+ T
CD8+ T
DC
B
NK
Other
Other T
linked genes
2 4 6 8 10+
0
20,000
40,000
60,000
CA−Link+ peaks
f
Mode
Rheostat
Dual
Switch
Density (×10−6)
g
Fraction
50%
40%
30%
20%
10%
0%
−1
−0.5
0
0.5
1
Distance to TSS (Mb)
h
PLS
pELS
dELS
CA−CTCF
CA−H3K4me3
CA−TF
CA
TF
Mixed
ENCODE4
Rheostat
Dual
Switch
0
1
2
3
4
5
Enrichment